

Linear Correlations in R

Correlation

A bivariate correlation is a statistical measure that expresses the extent to which two variables are related. It is a common tool for describing simple relationships without making a statement about cause and effect.

Correlations range in value from -1 to +1. A value of 0 indicates no correlation.

Pearson's (product moment correlation) is used to measure the strength of a **linear correlation** between two variables.

The sample correlation r_{xy} can be used to estimate the population Pearson's correlation ρ_{xy}

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_j (x_j - \bar{x})^2 \sum_j (y_j - \bar{y})^2}}$$

Evans (1996) (<https://doi.org/10.2466/pms.1996.82.3.988>) gives the following guidelines for strengths of correlations:

- $0.0 \leq |r| < 0.2$ "very weak"
- $0.2 \leq |r| < 0.4$ "weak"
- $0.4 \leq |r| < 0.6$ "moderate"
- $0.6 \leq |r| < 0.8$ "strong"
- $0.8 \leq |r| \leq 1.0$ "very strong"

Other available correlation measures are Kendall's tau or Spearman's rho (both rank correlation coefficients, which don't have the same assumptions about linearity).

The important statistical functions in R for calculating the correlation between variables are `cor` and `cor.test`. The function `cor` will just return the value of the correlation coefficient. The function `cor.test` will provide further statistical information, including p-values for judging whether or not the correlation is statistically significantly different from zero.

An example implementation:

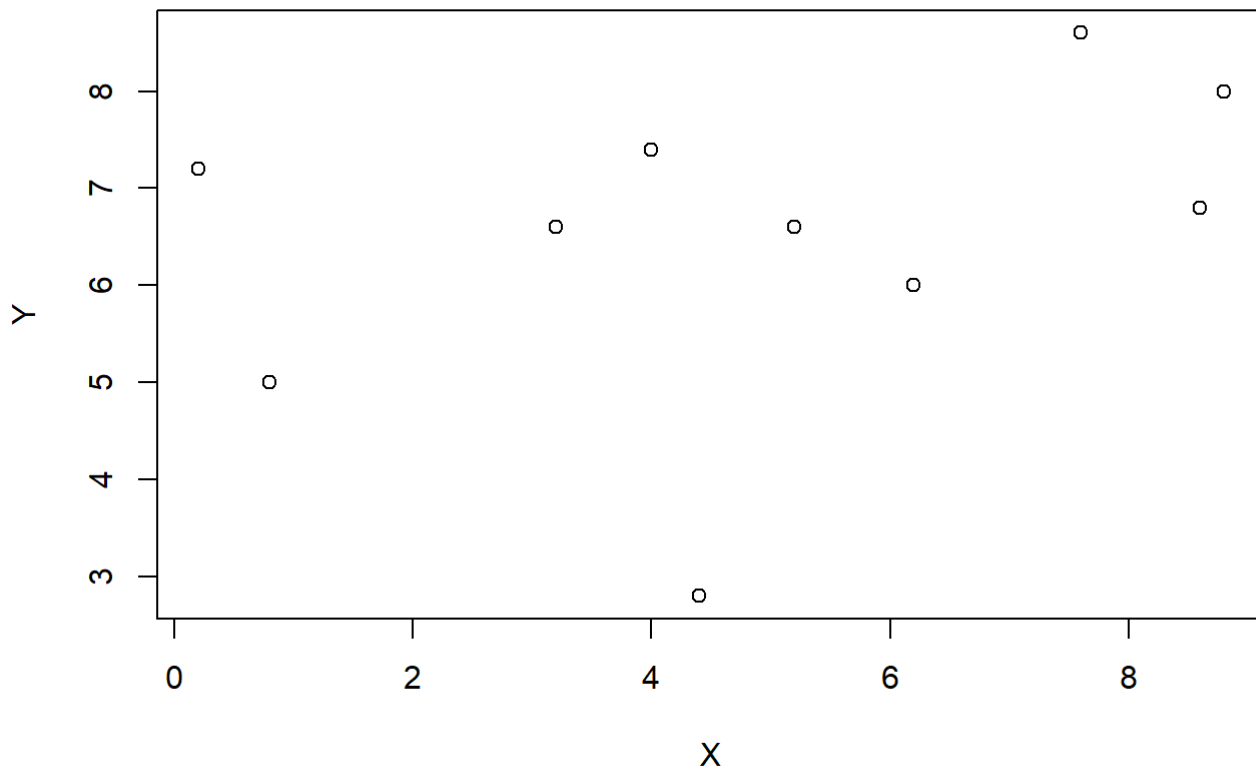
Working in my Statistical Inference working directory (change using Session > Set Working Directory > Choose Directory), I have created a subfolder called "DEWIS" for these online tests, where I have also stored this rmd file.

Load the data:

```
Example1<-read.csv("correlation.csv")
```

Basic plot:

```
with(Example1,plot(X,Y))
```



Question: on the basis of this graph, do you think you have an obvious linear relationship between X and Y ?

Find the correlation coefficient

```
cor(Example1$X,Example1$Y)
```

```
## [1] 0.3554675
```

We could also embed this into our code, reporting that the correlation coefficient between X and Y here is 0.355 (``r` round(cor(Example1$X,Example1$Y),3)``) (rounding to three decimal places).

If we wanted to conduct a formal statistical hypothesis test, then we would need to use the function `cor.test`

```
cor.test(Example1$X,Example1$Y)
```

```
##
## Pearson's product-moment correlation
##
## data:  Example1$X and Example1$Y
## t = 1.0757, df = 8, p-value = 0.3134
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.3532117  0.8049390
## sample estimates:
##      cor
## 0.3554675
```

Task: investigate the help file associated with this function to see what options are available.

Suppose we want to be able to extract parts of the output of the `cor.test` function. Then we save the results as an object.

```
rXY<-cor.test(Example1$X,Example1$Y)
names(rXY)
```

```
## [1] "statistic"  "parameter"  "p.value"    "estimate"   "null.value"
## [6] "alternative" "method"     "data.name"  "conf.int"
```

In this way we can pull out the estimate of r : `0.3554675 ( `r` rXY$estimate `)`, $df = 8$ (``r` rXY$parameter``), $p = 0.3134434$ (``r` rXY$p.value``) for reproducible reporting.

Think about how you may use `ifelse` statements to report in a reproducible:

1. If you have found a statistically significant linear correlation
2. The strength of the linear correlation detected.